Chapter 9: Project Resource Management

Information Technology Project Management, Ninth Edition

Note: See the text itself for full citations

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Learning Objectives (1 of 2)

- Explain the importance of good resource management on projects
- Define project resource management and understand its processes
- Summarize key concepts for managing people by understanding theories of motivation, influence, and power; how people and teams can become more effective; emotional intelligence; and leadership
- Discuss resource management planning and be able to create a human resource plan, project organizational chart, responsibility assignment matrix, and resource histogram
- Describe the process of estimating activity resources

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Learning Objectives (2 of 2)

- Discuss issues that are typically involved in resource acquisition, particularly as they involve resource assignments, resource loading, and resource leveling
- Assist in team development with training, team-building activities, and reward systems
- Explain and apply several tools and techniques to help manage a project team and summarize general advice on managing teams
- Summarize the process of controlling resources
- Describe how project management software can assist in project resource management
- Discuss considerations for agile/adaptive environments

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The Importance of Resource Management

- People determine the success and failure of organizations and projects
  - Most project managers agree that managing human resources effectively is one of the toughest challenges they face
- Proactive organizations are addressing human resource needs
  - Improving benefits
  - Redefining work hours and incentives
  - Finding future workers

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Keys to Managing and Leading People

- Psychologists and management theorists have devoted much research and thought to the field of managing and leading people at work
- Psychosocial issues that affect how people work and how well they work
  - Motivation theories
  - Influence and power
  - Highly effective people
  - Emotional intelligence
  - Leadership

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Motivation Theories

- Involving understanding what motivates people to do what they do
  - Help anyone who works or lives with others to understand themselves and others
- Intrinsic motivation causes people to participate in an activity for their own enjoyment
  - Example: some people love to read, write, or play an instrument because it makes them feel good
- Extrinsic motivation causes people to do something for a reward or to avoid a penalty
  - Example: some young children would prefer not to play an instrument, but they do because they receive a reward or avoid a punishment for doing so
- Include
  - Maslow's Hierarchy of Needs
  - Herzberg's Motivational-Hygiene Theory
  - Pink's Drive
  - McClelland's Acquired-Needs Theory
  - McGregor's Theory X and Y

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Maslow's Hierarchy of Needs

- Abraham Maslow argued that humans possess unique qualities that enable them to make independent choices, thus giving them control of their destiny
  - Developed a hierarchy of needs which states that people's behaviors are guided or motivated by a sequence of needs
  - The order and sizes of the needs are significant
    - A person cannot consider a need without the one in the lower level being satisfied
    - Once a need is satisfied, it no longer serves as a motivator of behavior

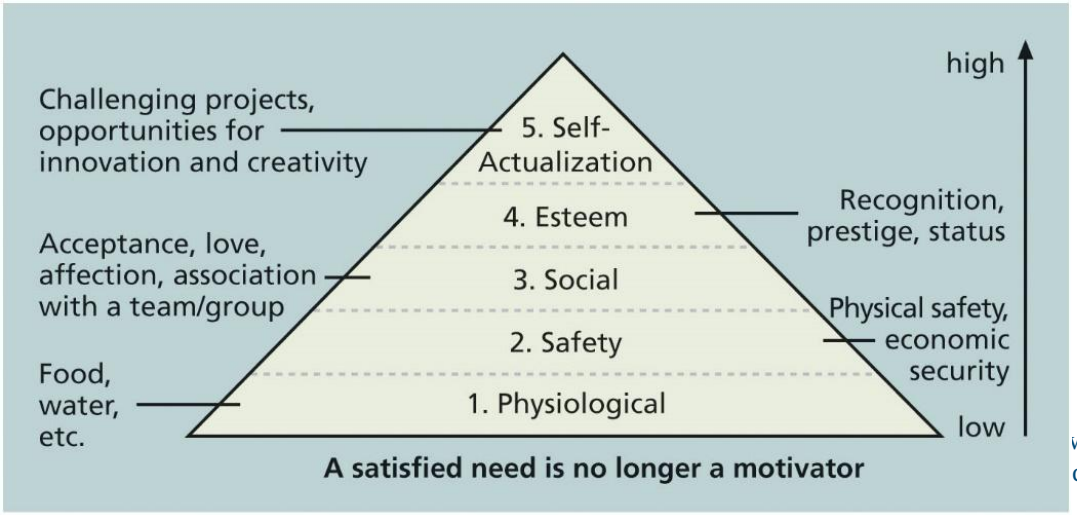


FIGURE 9-2 Maslow's hierarchy of needs

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Herzberg's Motivation-Hygiene Theory

- Frederick Herzberg distinguished between
  - Motivational factors: factors that cause job satisfaction (and motivate to work)
  - Hygiene factors: could cause job dissatisfaction if not present (but not motivate workers to do more)

Hygiene Factors	Motivators
Larger salaries and benefits	Achievement
Supervision quality	Recognition
More attractive work environment	Meaningful work
Computer or other required equipment	Responsibility
Job security	Advancement opportunities
Training	Personal growth and development

Table 9-1 Examples of Herzberg's hygiene factors and motivators

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Pink's Drive

- Daniel Pink's book called "Drive: The surprising truth about what motivates us"
- Pink suggests that managers focus on motivators
  - Autonomy: People are motivated when they have control over what they do, when they do it, how they do it, and with whom
    - E.g. flexible work schedules, self-managed projects, or freedom to experiment
  - Mastery: People like to improve their skills and master work-related activities
    - E.g. challenging work, opportunities for training, or time to practice and develop expertise
  - Purpose: People are most engaged when their work connects to a meaningful goal beyond personal gain
    - E.g. working for an organization with a clear mission or contributing to social impact projects



Drive: The surprising truth about what motivates us
https://www.youtube.com/watch?v=u6XAPnuFjJc

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McClelland's Acquired-Needs Theory

- David McClelland proposed that specific needs are acquired or learned over time and shaped by life experiences; everyone has these three needs, but one tends to dominate behavior
  - Need for Achievement (nAch)
    - Desire to excel, set and accomplish challenging goals, and seek personal success
    - Prefer tasks where they can take responsibility for outcomes, the risk is moderate (not too easy, not impossible), and they can receive feedback on performance
  - Need for Affiliation (nAff)
    - Desire for friendly relationships, social approval, and belonging
    - People high in nAff seek cooperation, avoid conflict, and prefer collaborative tasks
  - Need for Power (nPow)
    - Desire to influence, control, or have an impact on others
    - Two forms: Personalized power – seeking power for personal status or control; Socialized power – seeking power to achieve organizational or group goals.
- Measured by the Thematic Apperception Test (TAT) (https://www.utpsyc.org/TATintro/)

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McGregor's Theory X and Y (and Z)

- Douglas McGregor popularized the human relations approach to management in the 1960s
  - Theory X (Authoritarian, Micromanagement, Tight Supervision)
    - Assumes employees are naturally lazy, avoid work, seek security, show little ambition, and need to be controlled, directed, and threatened to perform
    - E.g. a manager believes employees will slack off unless constantly monitored and given strict rules
  - Theory Y (Participative decision making, Empowering)
    - Assumes employees are self-motivated, find work satisfying, seek responsibility, creativity, and opportunities for personal growth
    - E.g. a manager provides autonomy, encourages collaboration, and trusts employees to take initiative
  - Theory Z (Added in 1981 by William Ouchi)
    - Based on Japanese management practices
    - Emphasizes long-term employment, collective decision-making, and loyalty
    - Employees are motivated by a strong company culture, trust, teamwork, and holistic concern for employees (including family and well-being).
    - E.g. companies invest in lifetime employment and group harmony to motivate employees

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Influence and Power (1 of 3)

- Project managers must understand what types of influence and power to use in different situations

- Influence is the ability to alter other people's perceptions of a situation and makes use of positive language and subtle behaviors to encourage the desired outcome
- Thamhain and Wilemon's types of influence
  - Authority: legitimate hierarchical right to issue orders
  - Assignment: ability to influence a worker's later work assignments
  - Budget: ability to authorize others' use of discretionary funds
  - Promotion: ability to improve a worker's position
  - Money: ability to increase a worker's pay and benefits
  - Penalty: ability to cause punishment
  - Work challenge: ability to assign work that capitalizes on a worker's enjoyment of doing a particular task
  - Expertise: perceived special knowledge that others deem important
  - Friendship: ability to establish friendly personal relationships between the project manager and others

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Influence and Power (2 of 3)

- Ways to influence
  - When project managers use work challenges and expertise to influence people, projects are more likely to succeed
  - Projects are more likely to fail when project managers rely too heavily on authority, money, or penalty

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Influence and Power (3 of 3)

- Power is the ability to change the behavior of others and make them do things that they might not do otherwise
  - Power is much stronger than influence, because it is often used to force people to change their behavior
- French and Raven's types of power
  - Coercive
    - Punishment, threats, other negative approaches
    - Similar to Penalty category of influence
    - Correlates with project failure but can be effective in stopping negative behavior
  - Legitimate
    - Getting people to do things based on position of authority
    - Similar to Authority category of influence
    - Overemphasis correlates with project failure
  - Expert
    - Using personal knowledge and expertise to get people to change behavior
    - People will follow suggestions
  - Reward
    - Using incentives to induce people to do things, e.g. money, status, recognition, promotion, special work assignment
  - Referent
    - Based on person's own charisma, the person is held in very high regard, and people will follow

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Highly Effective People (1 of 2)

- Project managers can apply Covey's 7 Habits of Highly Effective People
  - Be proactive
    - Anticipating and planning for problems and inevitable changes
  - Begin with the end in mind
    - Focusing on what to be achieved, write mission statements
  - Put first things first
    - Schedule time for tasks based on importance rather than urgency, because they build long-term effectiveness
    - E.g. handling emails instantly, minor deadlines, interruptions vs. developing plans, building relationships with stakeholders, mentoring team members
  - Think win/win
    - When making decision and when resolving conflicts by developing new solutions that benefit all parties
  - Seek first to understand, then to be understood
    - Empathic listening or active listening – listening with the intent to understand other person, setting aside your personal interest
  - Synergize
    - Valuing differences in others to create synergy, creating collaborative results that are much better than a collection of individual efforts
  - Sharpen the saw
    - Renewing yourself physically, spiritually, mentally, and socially, and avoiding burnout

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Highly Effective People (2 of 2)

- Good project managers are empathic listeners
- Before you can communicate with others, you have to have rapport
  - Mirroring is a nonverbal technique where you subtly imitate another person's gestures, posture, tone, or speech patterns.
  - When done naturally (not forced), it creates a subconscious sense of similarity and comfort.
  - It helps build rapport faster because people tend to like those who seem like them.
  - E.g. if someone leans forward while talking, you also lean in slightly. If they speak calmly and slowly, you adjust your tone to match.
- IT professionals need to develop empathic listening and other people skills to improve relationships with users and other stakeholders

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Emotional Intelligence

- Howard Gardner's book *Frames of Mind: The Theory of Multiple Intelligences* introduced the concept of using more than one way to think of and measure human intelligence
  - Gardner suggested the need to develop both interpersonal intelligence (capacity to understand the motivations, intentions, and desires of others) and intrapersonal intelligence (capacity to understand oneself, one's feelings, and motivations)
    - Interpersonal intelligence example: empathic listening
    - Intrapersonal intelligence example: knowing that you have a high need for achievement
  - Emotional intelligence (EI) is knowing and managing one's own emotions and understanding the emotions of others for improved performance
    - Project manager uses EI when working on team building, collaboration, negotiation,and relationship development



https://greatergood.berkeley.edu/quizzes/ei_quiz/take_quiz
https://www.psychologytoday.com/us/tests/personality/emotional-intelligence-test

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Leadership

- A leader focuses on long-term goals and big-picture objectives, while inspiring people to reach those goals
- There is no one best way to be a leader
  - Most experts agree that the best leaders are able to adapt their style to needs of the situation
- Daniel Goleman's six leadership styles
  - Coercive (Commanding), Authoritative (Visionary), Affiliative, Democratic, Pacesetting, Coaching,
  - https://content.leadershipacademy.nhs.uk/aspce3/files/Leadership_that_gets_results_goleman.pdf



https://peopleinsight.co.uk/goleman-leadership-styles/

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Leadership Styles (1/2)

- Coercive (Commanding) "Do what I say"
  - Is effective in a turnaround situation, a natural disaster, or when working with problem employees.
  - Inhibits organization's flexibility and dampens employee's motivation.
- Authoritative (Visionary) "Come with me"
  - The leader states the goal but gives people the freedom to choose their own means of achieving it.
  - Works well when a business is adrift.
  - Is less effective when the leader work with a team who are more experienced.
- Affiliative "People come first"
  - Useful for building team harmony or increasing morale when the team has experienced conflict, stress, or difficult circumstances.
  - Focus on praise can allow poor performance to go uncorrected.
  - The leader rarely offers advice which often leaves employees in a quandary.

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Leadership Styles (2/2)

- Democratic
  - The leader may establish the overall goals or framework but encourages team members to share their opinions, participate in decisions, and take collective responsibility.
  - Creates a stronger sense of ownership and commitment and brings together diverse ideas and perspectives.
  - Sometimes resulting in endless meetings and confused employees who feel leaderless.
- Pacesetting
  - The leader sets high performance standards and exemplifies them himself.
  - Expects team members to work quickly, produce high-quality results, and meet the standards.
  - Positive impact on self-motivated and highly competent employees.
  - Other employees tend to feel overwhelmed by the leader's demands for excellence.
- Coaching
  - Focuses on developing and strengthening the capabilities of individuals and the team for current roles and future careers, rather than simply directing on immediate work-related tasks.
  - Works well when employees are aware of their weaknesses and want to improve, but not when they are resistant to changing their ways.

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Coercive (Commanding), Authoritative (Visionary), Affiliative, Democratic, Pacesetting, or Coaching?

Which leadership style are you using?





To improve and modernize software development process, you tell the team that you decide to overhaul the process by using agile development and automated testing. These are radically different from waterfall development and manual testing that the team are used to. You believe that these changes will make a real difference to the process and produced software.



You have just taken over the position from the previous project manager who was proficient but also did not care whose feelings she hurt. Now the team's morale is low. So you decide to focus on your team's emotional needs before doing any work. You let everyone open up about the problems and how they feel. After two meetings, the atmosphere in the team is better.



Although the New Year's holiday is coming up, your team has to release a new software version before the holiday. You think the team can handle the pressure, and if they can meet their performance goals, they will be rewarded with a great bonus. So you ask the team to work extra hours. You also work extra hours yourself, and help anyone who falls behind.

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Coercive (Commanding), Authoritative (Visionary), Affiliative, Democratic, Pacesetting, or Coaching?

Which leadership style are you using?





A new developer on the team is having trouble fitting in to his new role. The company requires employees to work at the office, but the new developer misses the freedom of remote working as he did at his old job. You help him see that coming to the office allows him to bond with colleagues who could turn into strategic allies in the future. You also encourage him to use the company's library to learn the skills he needs for a promotion.



The project sponsor needs to compress the schedule. The situation affects the whole team as lots of work still remain. You call a meeting with your team and explain the situation. You ask them how their tasks could be changed to make the project finish by the new end date.



Critical defects are found in the software. In a meeting, the team are constantly arguing and try to blame each other. You realize that nothing is going to get done unless someone takes charge. You start issuing orders for the defects to get fixed. Quickly, the room quiets down and you outline what needs to happen within the next few hours.

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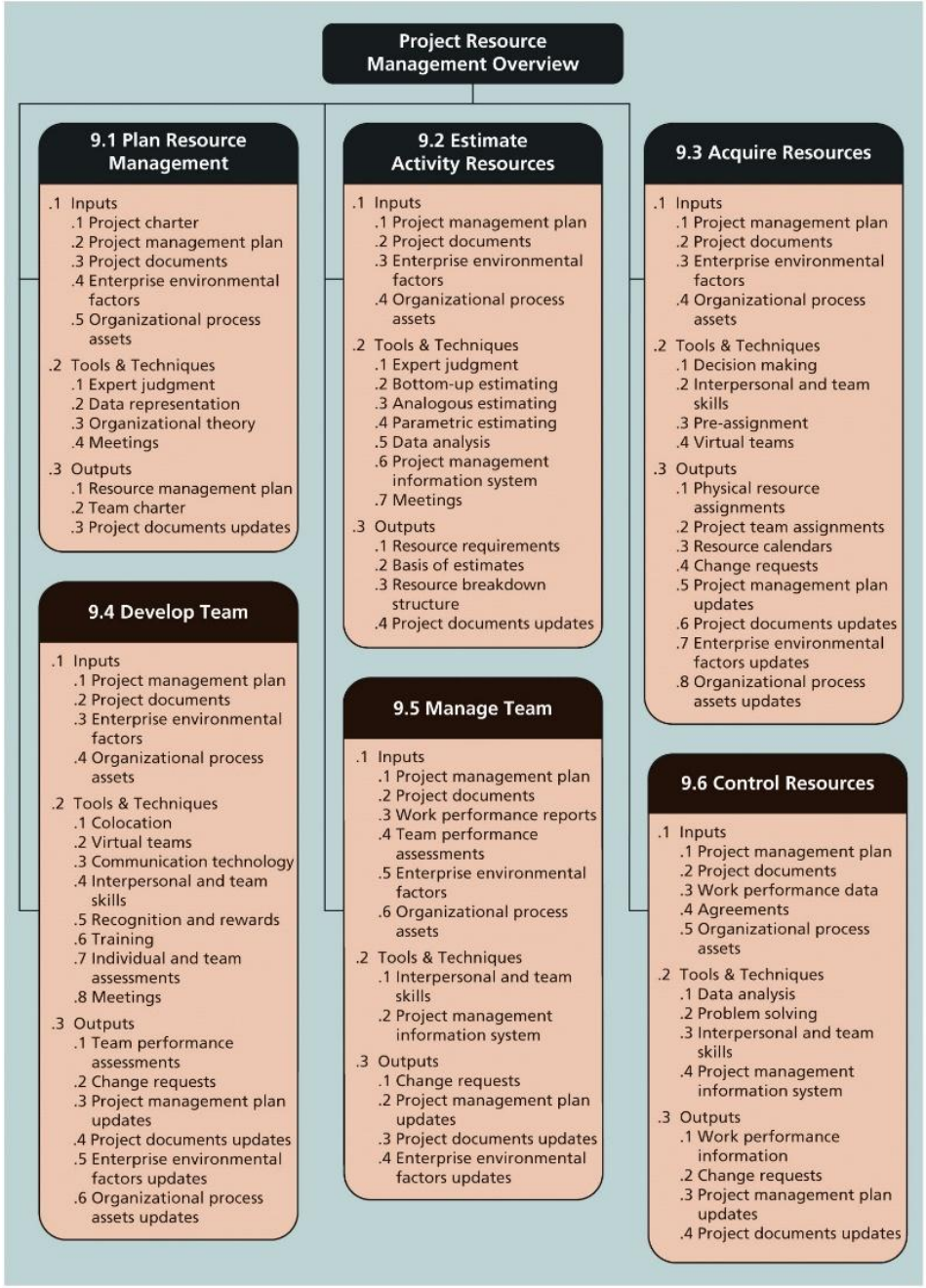
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Project Resource Management

- Making the most effective use of the human and physical resources involved with a project
- Project resource management processes

1. Planning resource management
   - Deciding how to estimate, acquire, manage, and use project resources
2. Estimating activity resources
   - Estimating human and physical resources needed to complete project work
3. Acquiring resources
   - Obtaining team members, facilities, equipment, materials, supplies, and other resources as needed
4. Developing the project team
   - Building individual and group skills
5. Managing the project team
   - Tracking team member performance, motivating team members, providing feedback, resolving issues and conflicts, coordinating changes
6. Controlling resources
   - Ensuring that physical resources are available as planned, monitoring planned versus actual resource utilization, taking corrective actions when needed

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FIGURE 9-1 Project resource management overview

1. Planning Resource Management

- To develop a resource management plan, identify and document project resources, roles, responsibilities, skills, and reporting relationships
  - Can be separated into a human resource management plan and a physical resource management plan
- Contents include:
  - Project organizational charts
  - Responsibility assignment matrixes
  - Staffing management plan and resource histograms
  - Team charters

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Project Organization Chart

- May have many types of structures (functional, matrix, etc. See Chapter 2)
- Created by project manager, top management, and team members
  - Smaller projects usually do not have deputy project managers or subproject managers

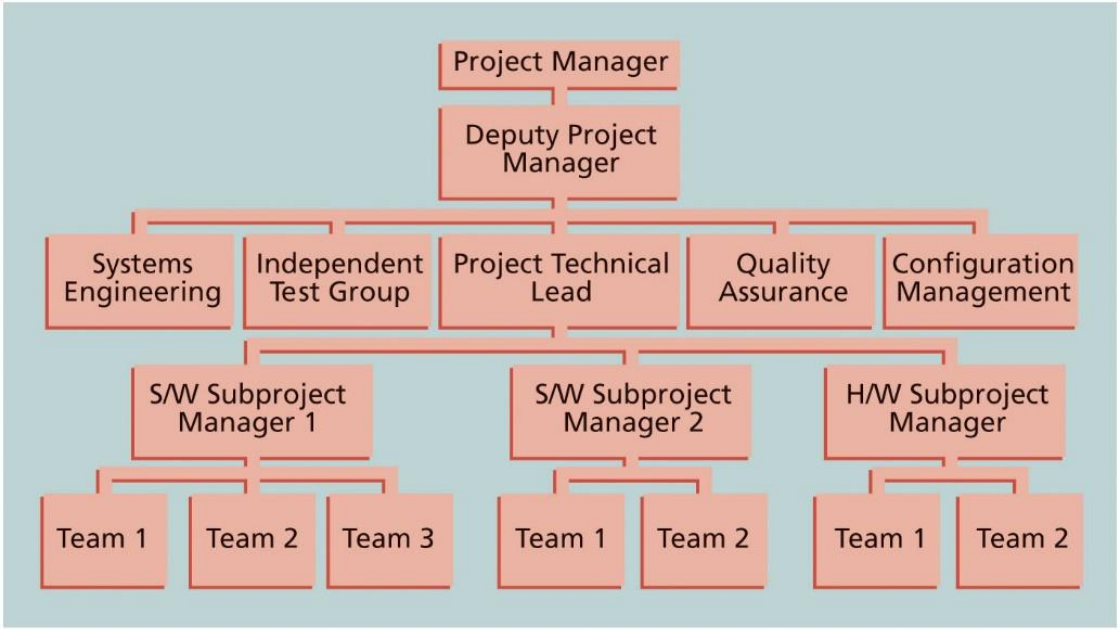
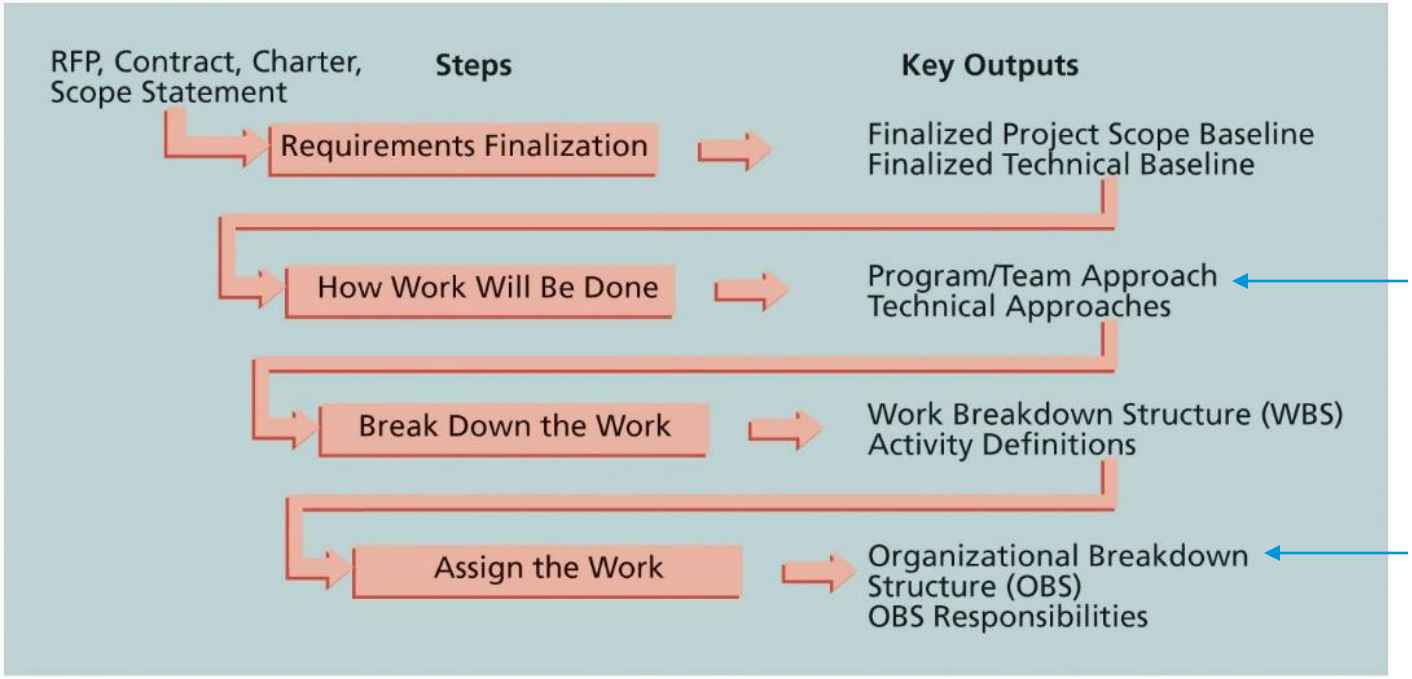


FIGURE 9-3 Sample organizational chart for a large IT project

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Work Definition and Assignment Process

- Carried out during the starting phase of the project and is iterative



Any outsourcing or subcontracting? Product- or phase-oriented work breakdown? Self-developed or packaged software?

Organization chart that is broken down into specific units within departments

FIGURE 9-4 Work definition and assignment process

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Responsibility Assignment Matrix (RAM)

- Map work of the project in WBS to responsible and performing people in OBS
  - People in OBS can be organizations, teams, individuals, depending on desired level of detail
  - For small projects, assign individual to WBS activities

WBS activities →

OBS units ↓

	1.1.1	1.1.2	1.1.3	1.1.4	1.1.5	1.1.6	1.1.7	1.1.8
Systems Engineering	R	R P					R	
Software Development			R P					
Hardware Development				R P				
Test Engineering	P							
Quality Assurance					R P			
Configuration Management						R P		
Integrated Logistics Support							P	
Training								R P

R = Responsible organizational unit
P = Performing organizational unit

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FIGURE 9-5 Sample responsibility assignment matrix (RAM)

RACI Chart

- A type of RAM used to define roles and responsibilities
  - Stakeholders can be included
- Key roles
  - Responsibility: Who does the task?
  - Accountability: Who signs off on the task or has authority for it?
  - Consultation: Who has information necessary to complete the task?
  - Informed: Who needs to be noticed of task status and results?

	Car Owner	Shop Owner	Mechanic	Parts Supplier
Pay for parts and services	A, R	C		
Determine parts and services needed	C		A, R	C
Supply parts		C	C	A, R
Install parts	I	A	R	

Table 9-2 Sample RACI chart

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Staffing Management Plan

- Describes when and how people will be added to the project team and taken off
  - Types of people, e.g. Java programmer, business analysts, technical writers etc.
  - Number of each type needed at a point in time, e.g. each month
  - How the resources are acquired, trained, rewarded, and reassigned after the project
  - Resource histogram: number of resources assigned to a project over time

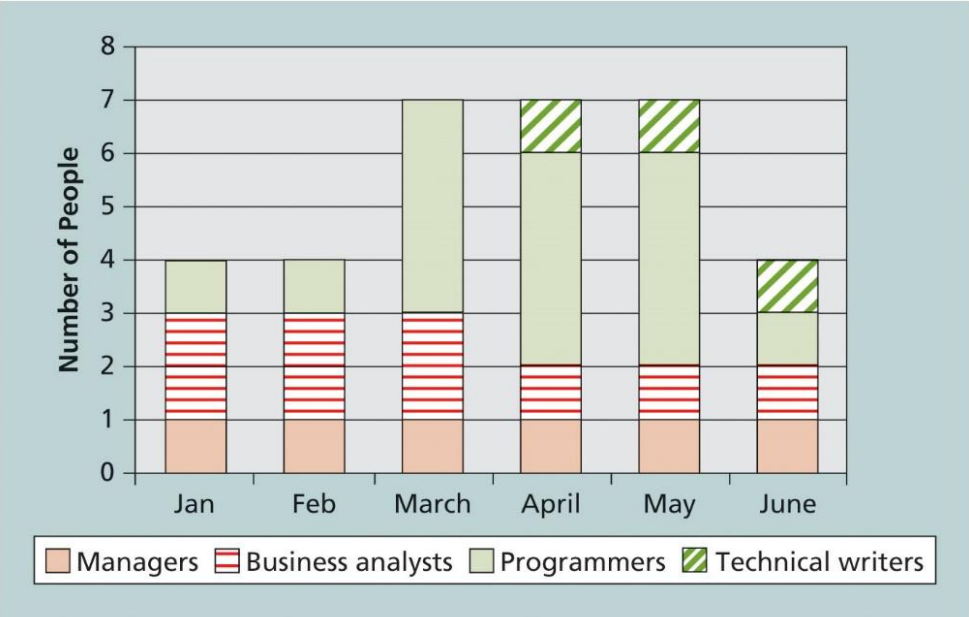


FIGURE 9-6 Sample resource histogram

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Team Charter

- Prepared by project team members
- Helps promote teamwork and clarify team communication
- Describes
  - Team values
  - Communication guidelines
  - Decision-making criteria and process
  - Conflict resolution process
  - Meeting guidelines
  - Team agreements
- See example of JWD in Chapter 3

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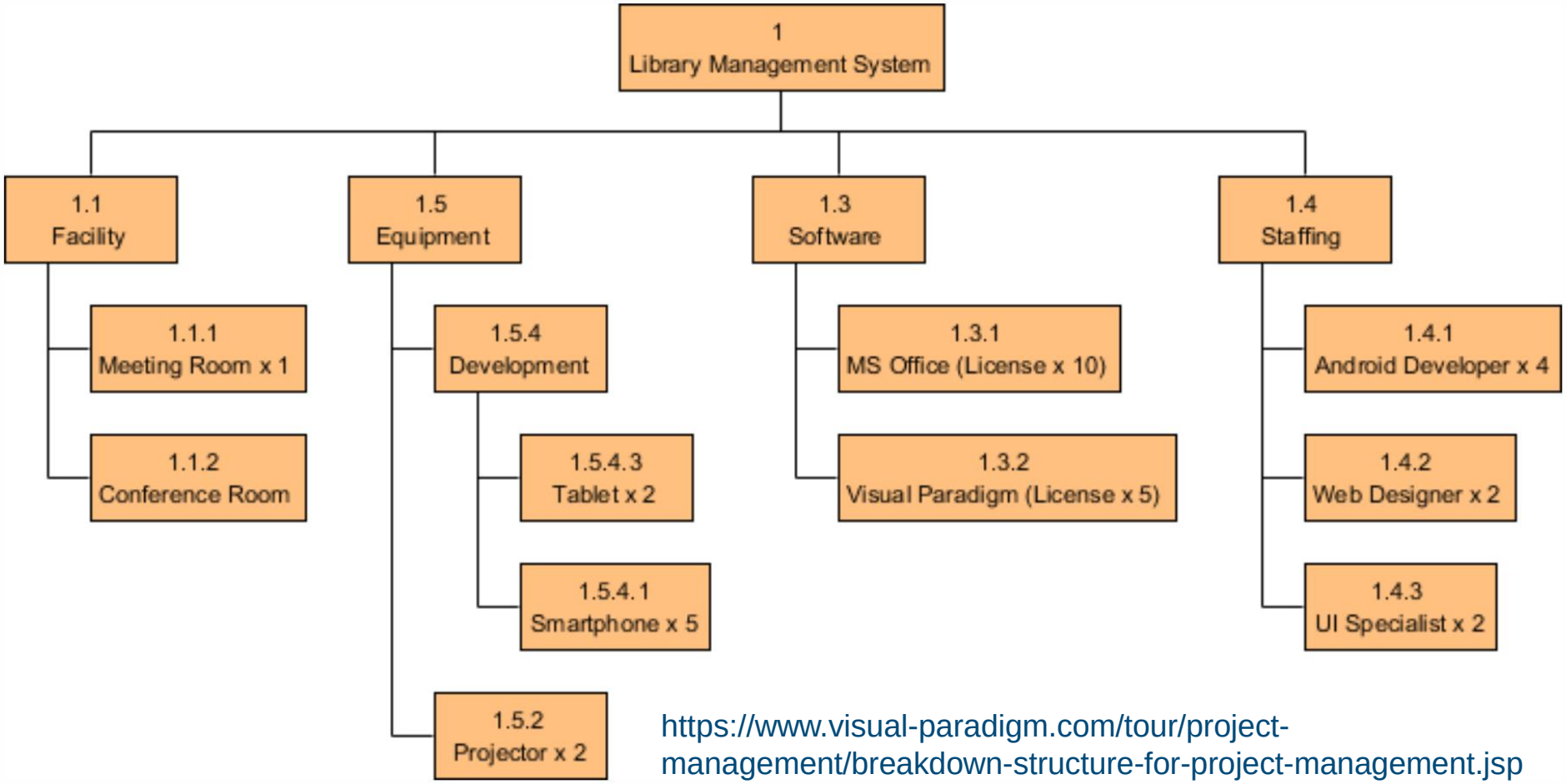
2. Estimating Activity Resources (1 of 2)

- Tools that can assist in resource estimating
  - Expert judgment
  - Various estimating approaches
    - Bottom-up
    - Analogous
    - Parametric
  - Data analysis
  - Project management software
  - Meetings
- May learn from past projects

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Estimating Activity Resources (2 of 2)

- Resource breakdown structure: hierarchical structure that identifies resources by category and types
  - Used in determining resource costs, acquiring resources, estimating activity durations etc.



https://www.visual-paradigm.com/tour/project-management/breakdown-structure-for-project-management.jsp

3. Acquiring Resources

- Recruiting team members
- Acquiring necessary physical resources, e.g. facilities, equipment, supplies
- Important topics
  - Resource assignment
  - Resource loading
  - Resource leveling

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Resource Assignment

- After developing resource requirements, project managers must work with other people in their organizations to
  - Assign human or physical resources to their projects or
  - Acquire additional resources needed for the project
- Inventory of employees' skills has to be maintained
  - If mismatch, the project manager, top management, HR managers, and others must work to address staffing and training needs
- It is very important to consider the needs of individuals and the organization when making recruiting and retention decisions

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Resource Loading (1 of 2)

- Amount of individual resources an existing schedule requires during specific time periods
  - Helps project managers develop a general understanding of the demands a project will make on the organization's resources and individual people's schedules
- Overallocation means more resources than available are assigned to perform work at a given time

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Resource Loading (2 of 2)

Percentage of available time (e.g. 100% = 8 hours per day)

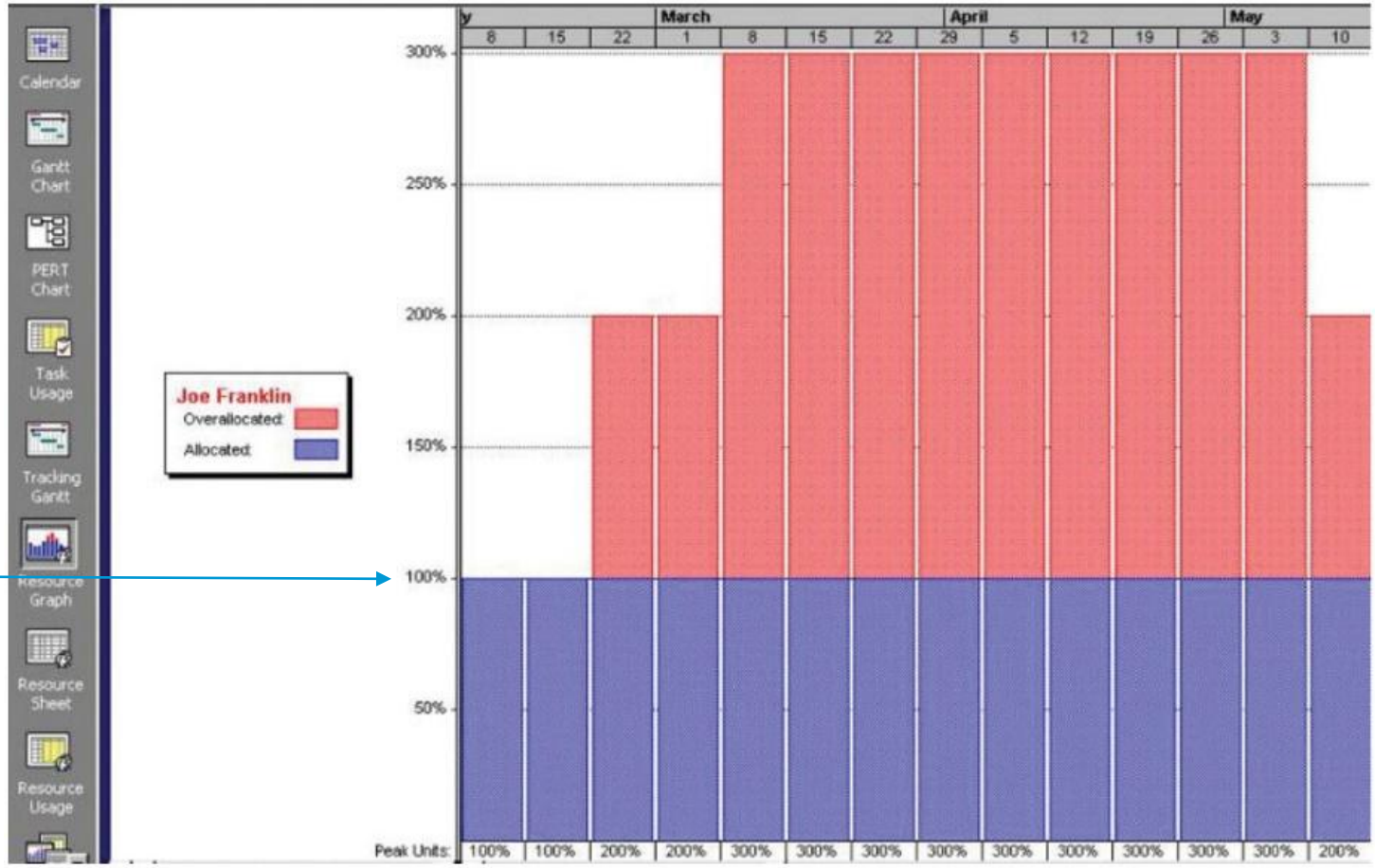


FIGURE 9-7 Sample histogram showing an overallocated person

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Resource Leveling (1 of 2)

- Technique for resolving resource conflicts by delaying tasks
  - Main purpose is to create a smoother distribution of resource usage
  - Try to shift tasks within their slack allowance to resolve overallocation or underallocation
  - Sometimes you need to delay project finish date though
- Benefits
  - When resources are used on a more constant basis, they are easier to manage
  - May enable project managers to use a just-in-time inventory type of policy for subcontractors or other expensive resources, instead of hiring more (which is usually more expensive)
  - Results in fewer problems for project personnel and accounting department
  - Often improves morale as people like to have stability in their jobs

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Resource Leveling (2 of 2)

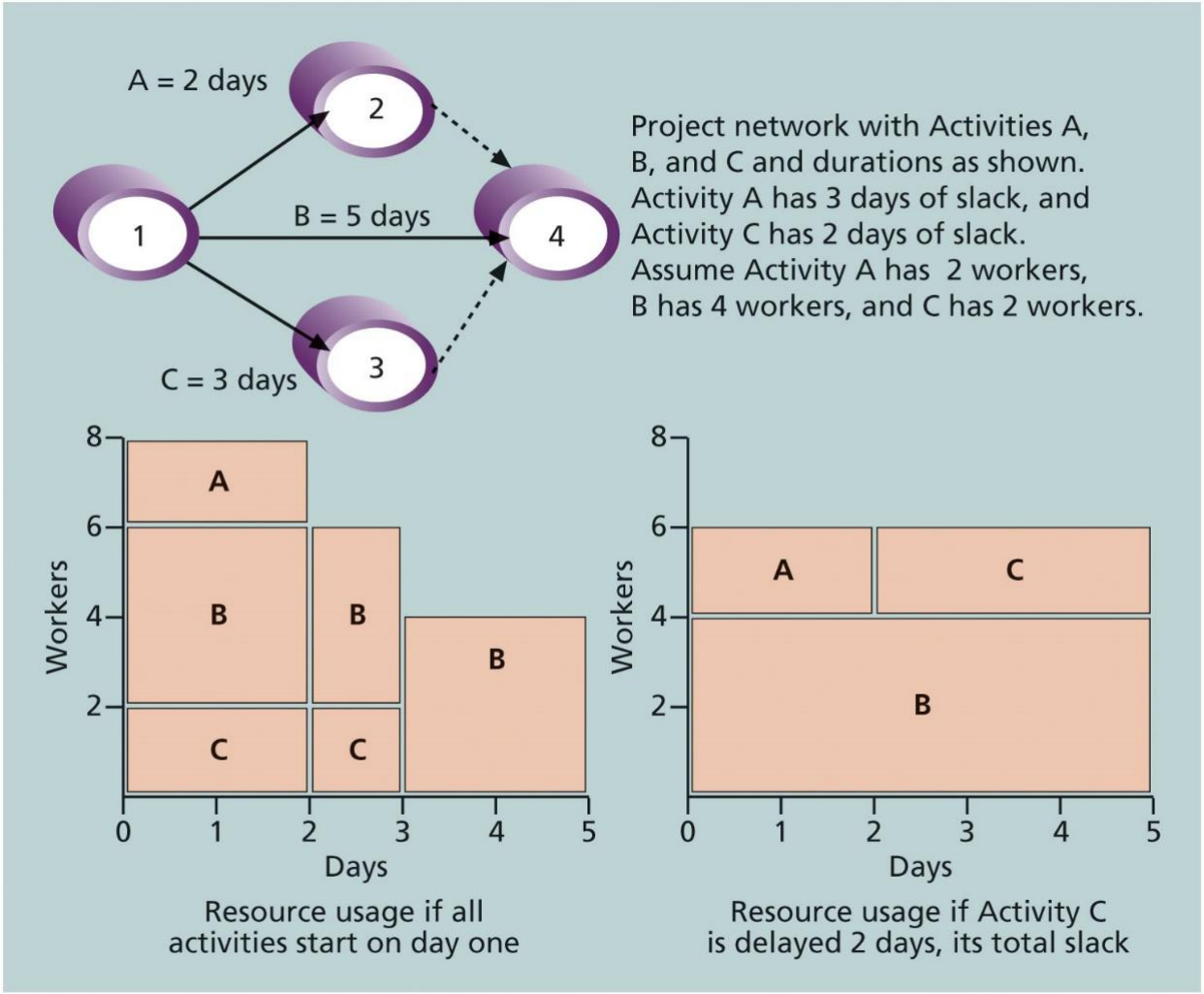


FIGURE 9-8 Resource leveling example

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4. Developing the Project Team (1 of 2)

- The main goal of team development is to help people work together more effectively to improve project performance
  - It takes teamwork to successfully complete most projects
- Tuckman model describes five stages of team development
  - Forming
    - Introduction of team members
  - Storming
    - Team members have different opinions for how the team should operate and test each other
    - Conflict within team
  - Norming
    - Team members develop a common working method
    - Collaboration and cooperation replace conflict and mistrust
  - Performing
    - Emphasis is on reaching the team's goal
    - Relationships are settled
  - Adjourning
    - Break-up of the team after it successfully reaches its goals

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Developing the Project Team (2 of 2)

- Training
  - Project managers often recommend that people take specific training courses to improve individual and team development
- Team-building activities
  - Physical challenges
  - Psychological preference indicator tools
    - Myers-Briggs type indicator
    - Disc profile
    - Social styles profile

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The Myers-Briggs Type Indicator (MBTI)

- Popular tool for determining personality preferences and helping teammates understand each other
- Four dimensions
  - Extrovert/Introvert (E/I)
    - Draw energy from other people (E) or from yourself (I)
  - Sensation/Intuition (S/N)
    - Prefer facts, details, and practical information (S) or be imaginative, ingenious, and attentive to hunches or intuition (N)
  - Thinking/Feeling (T/F)
    - Make decisions based on logic, analysis, and objectivity (T) or Make decisions based on values, empathy, and harmony (F)
  - Judgment/Perception (J/P)
    - Prefer structure, planning, and decisiveness (J) or Prefer flexibility, spontaneity, and adaptability (P)
- Project managers can benefit from knowing team members' MBTI profiles by adjusting management styles for each person



http://www.humanmetrics.com/

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DISC Profile (1 of 2)

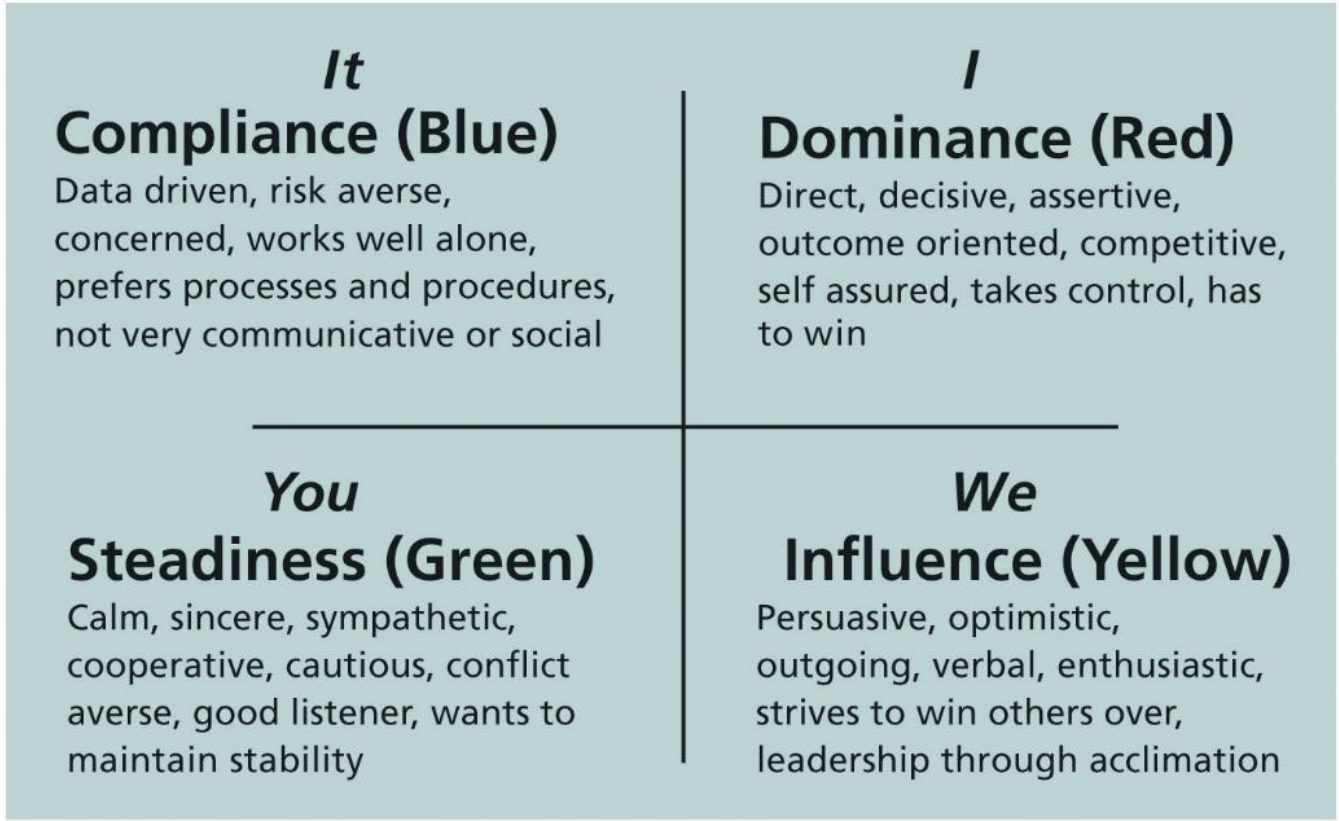
- William Moulton Marston described people's behavioral tendencies under certain situations in four dimensional model
- Each dimension is associated with a color and emphasis (I, We, You, or It)
  - Dominance
  - Influence
  - Steadiness
  - Compliance
- People in opposite quadrants may have problems understanding each other
- Project managers can determine the best way to interact with a person, and understand why certain people may have problems working with each other

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DISC Profile (2 of 2)

Focus on the task, details, facts, rules, systems

Focus on self, personal control, achievement, results



Focus on others, support, loyalty, helping people feel secure

Focus on relationships, group belonging, enthusiasm, connection

FIGURE 9-10 The DISC profile



https://personality.co/disc-test

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The Social Styles Profile (1 of 2)

- David Merril describes how people interact socially and at work by focusing on two dimensions
  - Assertiveness (the degree to which a person tends to ask vs. tell, or is passive vs. forceful)
  - Responsiveness (the degree to which a person shows emotions, controls emotions, is task-oriented vs. people-oriented)
- When crossing these two dimensions:
  - Drivers (Tell + Task-oriented)
  - Expressives (Tell + People-oriented)
  - Analyticals (Ask + Task-oriented)
  - Amiables (Ask + People-oriented)
- People in opposite quadrants may have problems understanding each other
- Project managers can determine the best way to interact with a person, and understand why certain people may have problems working with each other

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The Social Styles Profile (2 of 2)

- Precise, careful, detail-focused, logical.
- Value accuracy, data, structure.
- Strengths: Thorough, dependable.
- Weakness: Can be indecisive or overly critical.

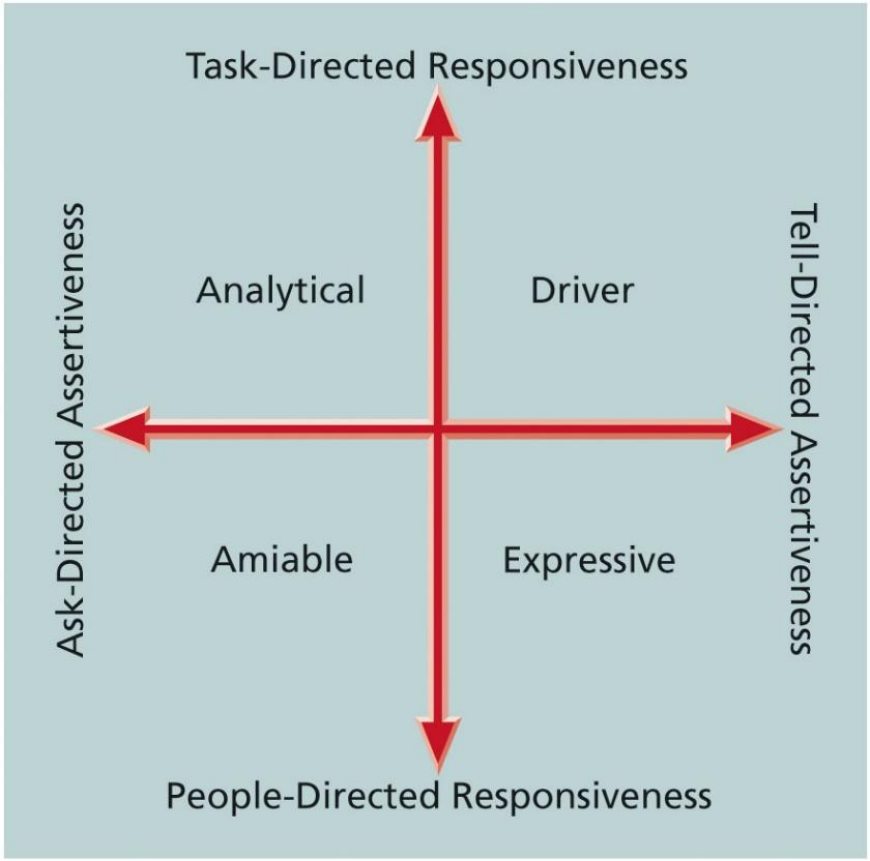


FIGURE 9-9 Social styles

- Direct, decisive, focused on results.
- Like control, efficiency, and speed.
- Strengths: Gets things done.
- Weakness: Can seem pushy or impatient.

- Supportive, cooperative, relationship-focused.
- Value trust, stability, and harmony.
- Strengths: Team player, empathetic.
- Weakness: May avoid conflict or delay decisions.

- Outgoing, enthusiastic, creative, persuasive.
- Value recognition, excitement, and relationships.
- Strengths: Visionary, motivating.
- Weakness: Can be impulsive or lack follow-through.



https://programs.changeosity.com/wp-content/uploads/2020/07/2.0-Prework_Social-Styles-Self-Assessment-20200701.pdf

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Reward and Recognition Systems

- Team-based reward and recognition systems can promote teamwork
  - Focus on rewarding teams for achieving specific goals
    - If management rewards teamwork, they will promote or reinforce the philosophy that people work more effectively in teams
- Project managers must continually assess their team's performance
  - When they find areas in which individuals or the entire team can improve, it's their job to find the best way to develop their people and improve performance

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5. Managing the Project Team

- Project managers must lead their teams in performing various project activities
- After assessing team performance and related information, the project manager must make several decisions
  - Changes to be requested
  - Corrective or preventive actions
  - Updates needed
- Tools and techniques to manage team
  - Interpersonal and team skills
    - Conflict management
    - Decision making
    - Emotional intelligence
    - Influencing
    - Leadership
  - Project management information systems

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Conflict Management (1 of 4)

- Blake and Mouton's five modes or strategies for handling conflicts
  - Confrontation (or problem-solving), Collaborating
    - Directly addressing the conflict by bringing the parties together to openly discuss differences and work toward a solution that satisfies both sides. (win-win)
    - Works best when the issue is important and long-term, and maintaining trust and collaboration is critical.
  - Compromise (or reconciliation)
    - Each party gives up part of what they want to reach an agreement that is acceptable (but not ideal) for everyone. (lose-lose or win some, lose-some)
    - Works best when a quick resolution is needed, and both sides have equal power.
  - Smoothing (or accommodating)
    - Emphasizing areas of agreement and downplaying areas of disagreement in order to reduce tension and maintain harmony in the relationship
    - One side may yield or sacrifice its own concerns for the sake of maintaining peace. (win-lose)
    - Works best when maintaining relationships is more important.

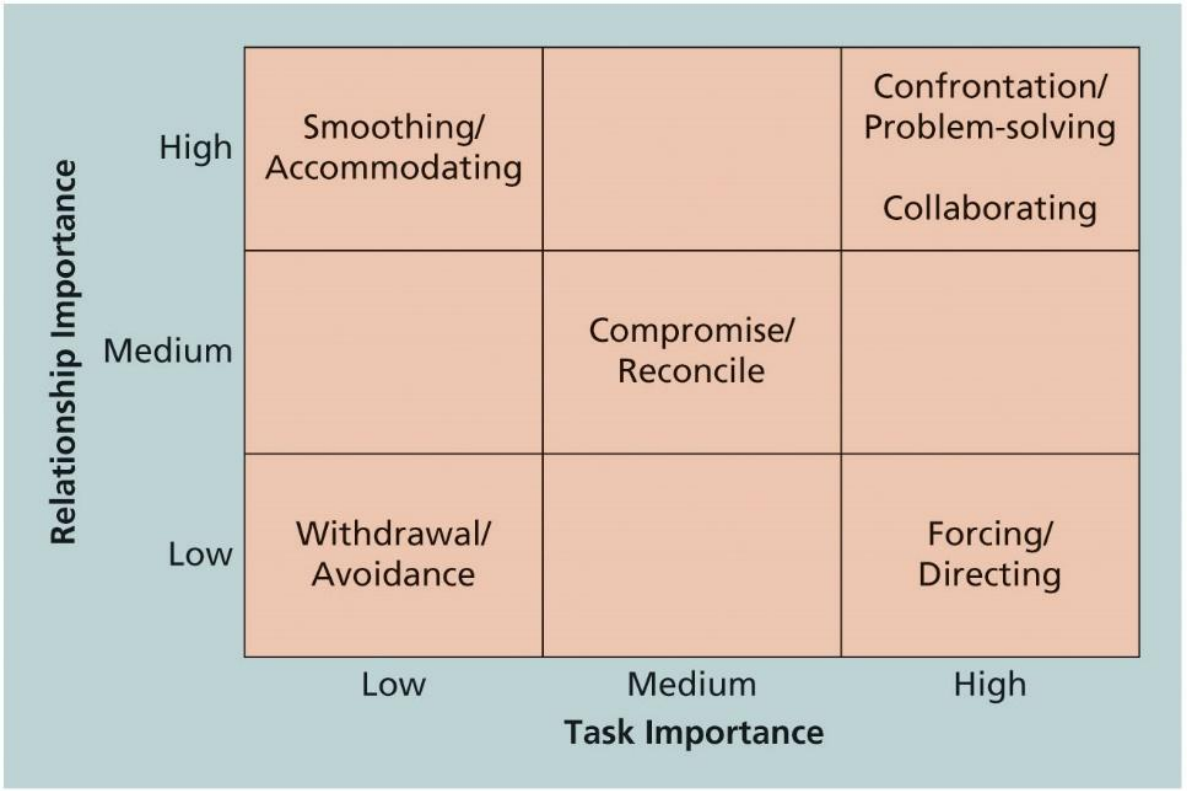
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Conflict Management (2 of 4)

- Blake and Mouton's five modes or strategies for handling conflicts
  - Forcing (or directing)
    - Using authority, power, or pressure to win a conflict at the expense of the other party. (win-lose)
    - Works best in emergencies where fast, decisive action is required, and when the issue is critical for business survival.
  - Withdrawal (or avoiding)
    - Removing yourself from the conflict instead of directly addressing it.
    - Postponing the conflict, leading to a temporary pause but no real resolution. (lose-lose)
    - Works best when the issue is trivial, a cool-down period is needed, or more information or time is needed before deciding.

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Conflict Management (3 of 4)



Source: *Kathy Schwalbe, An Introduction to Project Management, Sixth Edition* (September 2017)

FIGURE 9-11 Conflict handling modes

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Conflict Management (4 of 4)

- Conflict can be good
  - Conflict often produces important results, such as new ideas, better alternatives, and motivation to work harder and more collaboratively
    - On the contrary, team members may become stagnant and develop groupthink - conformance to the values or ethical standards of a group- if there are no conflicting viewpoints
  - Research suggests that task-related conflict often improves team performance, but emotional conflict often depresses team performance

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Confrontation, Compromise, Smoothing, Forcing, or Withdrawal?



Which conflict management strategy is/should be used?



The project manager manages a distributed team with team members located in different countries. Due to time zone differences, the team is in trouble finding a time that is acceptable to all team members for meetings. A perfect time for someone is always a very bad time for another.



The project manager thinks that the project should buy a printer from supplier A who supplied printers to previous projects with a good after-sales service. The analyst thinks that the project should try a new supplier B who offers a discount for toners. The project manager finally decides that supplier B is okay too.



For weeks, two designers disagree on the new UX design for the project. Since the design deadline is postponed, the project manager asks them to help in two other different projects for a week, allowing them to cool down and resume in a more collaborative mindset.

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General Advice on Managing Teams (1 of 2)

- Five dysfunctions of teams
  - Absence of trust
  - Fear of conflict
  - Lack of commitment
  - Avoidance of accountability
  - Inattention to results

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General Advice on Managing Teams (2 of 2)

- General advice
  - Be patient and kind with your team
  - Fix the problem instead of blaming people
  - Establish regular, effective meetings
  - Allow time for teams to go through the basic team-building stages
    - Don't expect team to work at the highest performance right away
  - Limit the size of work teams to three to seven members
  - Plan some social activities to help project team members and other stakeholders get to know each other better
  - Stress team identity
  - Nurture team members and encourage them to help each other
  - Acknowledge individual and group accomplishments
  - Take additional actions to work with virtual team members
    - Have face-to-face or phone meeting at the start
    - Make sure they can work effectively in virtual environment

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6. Controlling Resources

- Ensuring physical resources assigned to the project are available as planned
  - Also involves monitoring the planned versus actual resources utilization and taking corrective actions as needed

(Making effective use of team members is addressed under Managing the Project Team)

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Using Software to Assist in Resource Management

- Software (spreadsheet, project management software) can help in producing a simple responsibility assignment matrix or resource histograms
- Software includes several features related to human resource management
  - Assigning and tracking resources
  - Leveling resources
  - Resource usage reports
  - Overallocated resource reports
  - To-do lists

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Considerations for Agile/Adaptive Environments

- Teams are important on all types of projects, as is collaboration, problem solving, and knowledge sharing
- On agile projects, team members are usually fully dedicated to a single team
  - Relationships are based on trust, and collaboration is continuously improved using regular feedback
  - Delivery of usable product at end of sprint builds team confidence
  - Team is self-directing, using a Scrum master rather than project manager
  - Daily stand-up meeting increases frequency of interaction among team members
  - Co-location of users with development teams is a form of social needs satisfaction

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Chapter Summary

- Project resource management includes the processes required to make the most effective use of the people involved with a project
  - Planning resources
  - Estimating activity resources
  - Acquiring resources
  - Developing the team
  - Managing the team
  - Controlling resources

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